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Indian Statistical Institute
B. Statistical Data Science II (2025-26)
Mid Semester Examination 2026
Part-I

Paper: Mathematics IV
Time : 10.30 AM - 11.30 AM (1 Hrs.)

Date: 13/03/2026
Full Marks : 15

Note: Symbols used have their usual meaning. The figures in the margin indicate full marks.
*Answer any **THREE** from the following questions.* $5 \times 3 = 15$

1. (a) Draw the isoclines with their direction markers and sketch several solution curves of the differential equation $y' = y^3 - x^2$.
(b) Find the general solution of the following Cauchy-Euler equation

$$x^2 \frac{d^2 y}{dx^2} + 5x \frac{dy}{dx} + 3y = 0.$$

[2 + 3]

2. Let $G(x, y)$ be an integrating factor of the differential equation $M(x, y) dx + N(x, y) dy = 0$. If $F(x, y) = c$ represents the general solution, then prove or disprove that $G(x, y)\chi(F)$ is an integrating factor of the above differential equation. Here $\chi(F)$ is some integrable function of F . [5]

3. According to the 2001 and 2011 censuses, the population of a country was 550 million and 685 million, respectively. Assuming that the country has sufficient resources to support a maximum population of 1000 million, estimate the population of the country in 2031. [5]

4. (a) Consider the differential equation

$$(y^2 + 2xy) dx - x^2 dy = 0.$$

- (b) If f is an even function and y is an odd function then prove or disprove the following statement:
“ $-y(-x)$ and $y(-x)$ are solutions of the differential equation $y' = f(y(x))$ on $\mathbb{R}$ ”.

[5]

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Indian Statistical Institute
B. Statistical Data Science II (2025-26)
Mid Semester Examination 2026
Part-II

Paper: Mathematics IV

Date: 13/03/2026

Time : 11.45 AM - 12.45 PM (1 Hrs.)

Full Marks : 15

Note: Symbols used have their usual meaning. The figures in the margin indicate full marks.

*Answer any **THREE** from the following questions.*

$5 \times 3 = 15$

1. Find the general solution of the differential equation

$$\frac{d^4 y}{dx^4} + \frac{d^3 y}{dx^3} + \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = x^2 + 1.$$

[5]

2. Find the general solution of the following system of homogeneous linear differential equations

$$\frac{d}{dt} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7 & -1 \\ 4 & 3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

[5]

3. Examine the existence and uniqueness of the solution of the following initial value problem

$$y' = f(x, y), \quad y(0) = 0,$$

$$\text{where } f(x, y) = \begin{cases} 0, & \text{if } x = 0, |y| \leq 1, \\ 2x, & \text{if } 0 < |x| \leq 1, -1 \leq y < 0, \\ 2x - \frac{4y}{x} & \text{if } 0 < |x| \leq 1, 0 \leq y \leq x^2, \\ -2x, & \text{if } 0 < |x| \leq 1, x^2 \leq y \leq 1. \end{cases}$$

Using Picard's method of successive approximations find the n -th iteration $y_n(x)$ of the above initial value problem and discuss the convergence of the sequence $\{y_n\}$. [3 + 2]

4. Let $\phi_1(x)$ and $\phi_2(x)$ be two solutions of

$$y'' + a(x)y' + b(x)y = 0,$$

where $a(x)$ and $b(x)$ are continuous on an interval I .

- (a) Prove that the Wronskian $W(\phi_1, \phi_2)$ of $\phi_1(x)$ and $\phi_2(x)$ is either identically zero or nowhere zero on I .
(b) If $\phi_1(x)$ and $\phi_2(x)$ are linearly independent then show that the zeros of these functions are distinct and occur alternately.

[3 + 2]